Analysing the Economic Drivers of Underemployment in Sri Lanka: Findings from the 2015–2025 Period

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***Abstract*—Underemployment in Sri Lanka, encompassing time-related and qualification-based forms, remains a heavily under-analysed dimension of labour market slack. While headline unemployment stays near 3.8%, quarterly LFS micro-data reveals crisis peaks of 32.6% (2020 Q2) and 31.1% (2021 Q2). We investigate macroeconomic drivers over 2015–2025 via structural break detection (Zivot-Andrews, Bai-Perron PELT), Granger causality, ARDL bounds testing, and XGBoost with SHAP interpretability. GDP contraction and informal employment emerge as dominant SHAP drivers. ARDL bounds testing (n=36 quarterly) confirms long-run cointegration at 1% (F=7.25); the ECM speed-of-adjustment λ=-1.087 (p=0.003) indicates rapid but mildly overshooting quarterly correction. Sensitivity analysis confirms strong convergence of driver rankings across hours-based and qualification-based definitions ($\rho = 0.893$, $p = 0.007$). Gender-disaggregated SHAP reveals distinct driver structures by sex, with a statistically persistent female excess (Wilcoxon $p = 0.001$). The 2022 sovereign default structurally altered GDP and inflation elasticities, confirming underemployment as a far more sensitive labour market barometer than headline unemployment.**

***Index Terms*—Underemployment, Sri Lanka, ARDL, XGBoost, SHAP, Structural Breaks, Gender Disaggregation, Sensitivity Analysis**

I. Introduction

Sri Lanka's 2022 sovereign default, marked by foreign exchange collapse and inflation exceeding 70% [1], exposed a core flaw in relying on aggregate unemployment as the primary labour indicator. With unemployment stable at 3.8–5.5% throughout, systemic underutilisation was concealed. Two forms of underemployment surged: **time-related underemployment** (workers below 40 hrs/week seeking more hours), reaching 32.6% in 2020 Q2 and 31.1% in 2021 Q2 in quarterly series [2]; and **qualification-based underemployment** (workers below their educational level), driven by graduate mismatches and public sector freezes.

Despite being the primary driver of post-recession wage suppression [3], underemployment is absent from Sri Lanka's labour literature, which stops at 2020 Q4 [4]. We address four research questions: **RQ1** how underemployment evolved 2015–2025; **RQ2** which indicators associate most strongly; **RQ3** whether 2022 structurally altered these relationships; and **RQ4** evidence for long-run equilibrium.

This study is, to our knowledge, the first to: (1) apply SHAP-based interpretability to Sri Lankan underemployment across the complete 2015–2025 crisis arc; (2) simultaneously incorporate remittances, agricultural output, part-time employment, and discouraged workers as predictors; (3) conduct gender-disaggregated SHAP analysis; and (4) test structural breaks across 11 macro and labour series using Zivot-Andrews and Bai-Perron PELT. These directly address the three limitations of Pabasara and Silva [4]: focus on aggregate unemployment, analysis terminating before the sovereign default, and linear VECM specifications unsuited to crisis-induced nonlinearities.

II. Related Work

Bell and Blanchflower [3] develop an Underemployment Index showing that hours deficits linger long after unemployment recovers—mirroring Sri Lanka's 3.8% unemployment in 2023 amid persistent underemployment. The ILO 19th ICLS Resolution [5] formalises time-related and qualification-based underemployment as necessary components of a complete labour underutilisation account in high-informality economies. Haider [6] links underemployment spells to long-term earnings instability, while Autor [7] connects qualification mismatch to structural labour market polarisation relevant to Sri Lanka's growing graduate overhang.

The OECD [8] documents a 27% average underemployment increase across 29 post-crisis countries, with the sharpest spikes in economies undergoing fiscal consolidation—a pattern Sri Lanka follows. Flek and Mysikova [9] show that low-skilled workers cycle repeatedly through underemployment, consistent with Sri Lanka's post-2022 hysteresis. Chowdhury [10] attributes persistence to weak aggregate demand and informality across South and Southeast Asia; Islam and Verick [11] identify delayed fiscal transmission into hours-based slack consistent with our 2–3 quarter TLCC lags.

For Sri Lanka, Pabasara and Silva [4] confirm Okun's Law on aggregate unemployment via VECM (2005–2020 Q4) but do not address underemployment. Dissanayake [12] calls for crisis-updated forecasts motivating this work. Liyanapathirana and Samaraweera [13] establish agriculture as the primary

source of seasonal inactive hours, supporting inclusion of the Agricultural Output Index.

III. Data and Methodology

A. Dataset and Variables

We construct an annual panel (n=10, 2015–2024) and a quarterly dataset (n≈40, 2015 Q1–2024 Q4). Table I summarises all ten variables. The target variable is *time-related underemployment* from the DCS Quarterly Labour Force Survey [2]. Predictors cover demand (GDP Growth, Inflation), labour market structure (Informal Employment, Youth LFPR, Part-time Employment, Discouraged Workers), and external/sectoral channels (Remittances, Agricultural Output Index, Real Exchange Rate).

The Real Exchange Rate is constructed by splicing CBSL (2015–2020) and FRED DEXSLUS [1], [14] (2021–2024) nominal series and deflating with bilateral CPIs:

$$\mathrm{RER}_t = e_t^{\mathrm{nom}} \times \frac{\mathrm{CPI}_t^{\mathrm{USA}}}{\mathrm{CPI}_t^{\mathrm{LKA}}} \tag{1}$$

The 2022 annual LFS reports underemployment at an implausibly low 2.3% during a year of sovereign default; we use quarterly LFS data as the primary series and impute the annual value as the mean of 2021 and 2023 figures, flagged explicitly. Missing values are handled via KNN (isolated gaps) and MICE [15] (block gaps) imputation, with Rubin's rule pooling over m=5 datasets. Annual-only predictors (remittances, agricultural output) are temporally disaggregated to quarterly frequency using the Denton-Cholette method [16].

TABLE I
Variable Summary (Annual 2015–2024)

Variable	Source	Unit
Underemployment Rate	DCS LFS	%
GDP Growth Rate	CBSL/World Bank	% YoY
Inflation Rate (CPI)	CBSL	% YoY
Exchange Rate	CBSL + FRED	LKR/USD
Youth LFPR (15–24)	DCS LFS/ILO	%
Informal Employment	DCS LFS	% emp.
Agri. Output Index	FAO	Index
Remittances	World Bank	USD mn
Part-time Employment	ILO	% emp.
Discouraged Workers	DCS LFS	Number

B. Econometric Pipeline

Stationarity. ADF and KPSS tests [17] confirm most predictors are I(1); all become stationary in first differences.

Structural Breaks. Zivot-Andrews Model C [18] and Bai-Perron PELT are applied to all eleven series (Table II). Informal employment breaks significantly at 2017 (t=−6.02, p<0.01) and part-time employment at 2020 (t=−5.43, p<0.05), both predating the sovereign default.

Granger Causality and ARDL. On the quarterly dataset (n=36, maxlag=4), Agricultural Output Index is the only predictor clearing the Bonferroni-corrected threshold (F=11.91, p=0.0016). The ARDL(p, q) specification is:

TABLE II
Zivot-Andrews Structural Break Results, Model C. Critical value at 5% is −5.08.

Variable	Break	ZA stat	Sig.
Underemployment	2021	−5.47	**
GDP Growth	2019	−3.85	–
Inflation	2022	−17.62	***
Informal emp.	2017	−6.02	***
Part-time emp.	2020	−5.43	**
Youth LFPR	2017	−1.18	–
Exchange rate	2019	−2.28	–
Remittances	2022	−3.57	–
Agri. output	2020	−4.50	–
Discouraged wkrs	2020	−3.26	–
TRU female	2020	−3.65	–

***p<0.01, **p<0.05, – not sig.

$$\Delta U_t = \alpha_0 + \sum_{i=1}^{p} \phi_i \, \Delta U_{t-i} + \sum_{j=0}^{q} \boldsymbol{\beta}_j' \, \Delta \mathbf{x}_{t-j} + \lambda_1 U_{t-1} + \boldsymbol{\lambda}_2' \mathbf{x}_{t-1} + \varepsilon_t \tag{2}$$

where λ_1 and $\boldsymbol{\lambda}_2$ are tested via Pesaran's F-bounds procedure [19]. AIC selects ARDL(3,1,1,2,0) with seasonal dummies. The bounds F-statistic is 7.25 (p=0.0008), exceeding the 1% upper critical value of 4.68 for k=5 predictors (Case III), confirming long-run cointegration. The ECM representation is:

$$\Delta U_t = \mu + \lambda \, \widehat{\mathrm{ECT}}_{t-1} + \sum_{i} \psi_i \, \Delta U_{t-i} + \sum_{j} \boldsymbol{\gamma}_j' \, \Delta \mathbf{x}_{t-j} + \varepsilon_t \tag{3}$$

C. Machine Learning and Interpretability

XGBoost [20] is fitted on an 80/20 temporal split with all nine predictors. SHAP [21] decomposes each prediction into additive feature contributions:

$$\hat{f}(\mathbf{x}) = \phi_0 + \sum_{i=1}^{M} \phi_i(x_i) \tag{4}$$

where ϕ_i is the Shapley value of feature i. SHAP values are cross-validated against OLS coefficient directions for consistency.

IV. Results

A. Underemployment Dynamics (RQ1)

Fig. 1 shows unemployment remaining flat at 3.8–5.5% while underemployment surges during 2020–2022. Quarterly extraction yields peaks of 32.6% (2020 Q2) and 31.1% (2021 Q2), with sustained 23–28% through 2022. Female underemployment consistently exceeds male (3.8% vs. 2.0% in 2015), with the gap widening at crisis peak (Fig. 2).

B. Structural Breaks and Interaction Effects (RQ3)

Bai-Perron PELT detects a dominant break at 2022, with co-occurring breaks in inflation (2021) and exchange rate (2021)

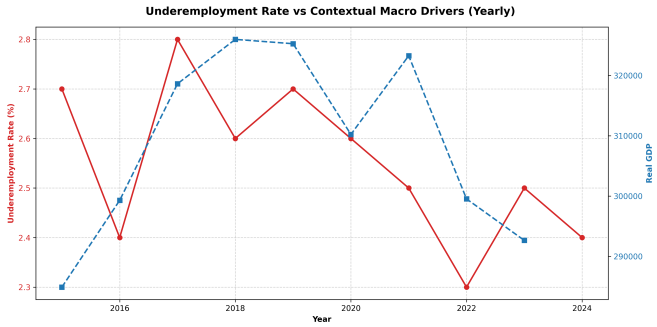

Fig. 1. Underemployment vs. macro indicators (2015–2024). Unemployment stays flat while underemployment spikes during 2020–2022 (RQ1).

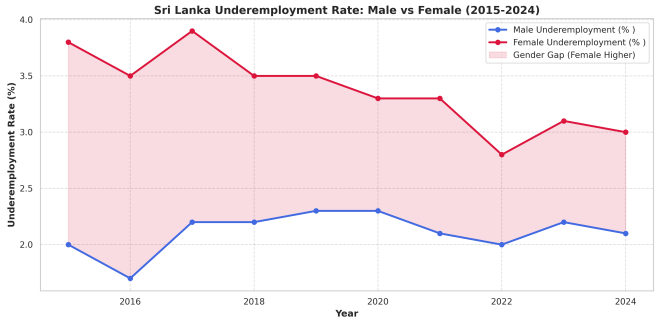

Fig. 2. Gender disaggregation (2015–2024). Female underemployment consistently exceeds male; gap widens at crisis peak.

preceding the labour market response by roughly one year. The post-break interaction model:

$$U_t = \alpha + \beta_1 G_t + \beta_2 D_t + \beta_3 (G_t \times D_t) + \varepsilon_t \qquad (5)$$

achieves R^2=0.495 for GDP and R^2=0.594 for inflation (Fig. 3), confirming structural amplification of macro elasticities post-2022. Informal employment breaking at 2017 suggests formalisation pressures predating the crisis.

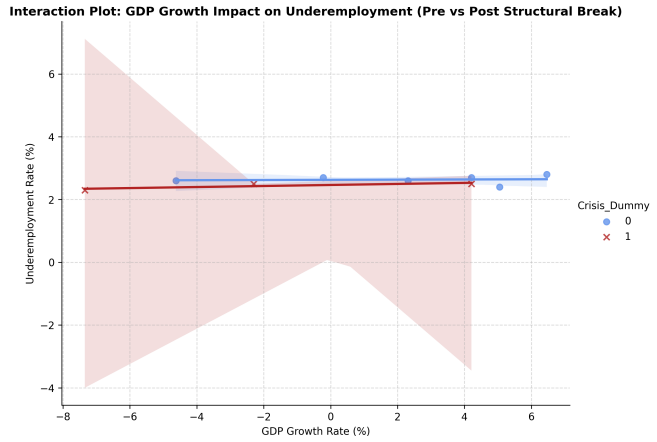

Fig. 3. GDP Growth × Crisis interaction. Steeper post-2022 slope confirms structural amplification (RQ3).

C. SHAP Feature Association (RQ2)

Fig. 4 shows **GDP Growth** as the primary demand-side driver; **Informal Employment** the second-ranked structural driver; **Youth LFPR** and **Remittances** capturing complementary channels. All top SHAP directions agree with OLS coefficient signs (cross-method validation). The 2022 SHAP waterfall (Fig. 5) shows GDP contraction and informal employment together accounting for the majority of the positive departure from the model baseline.

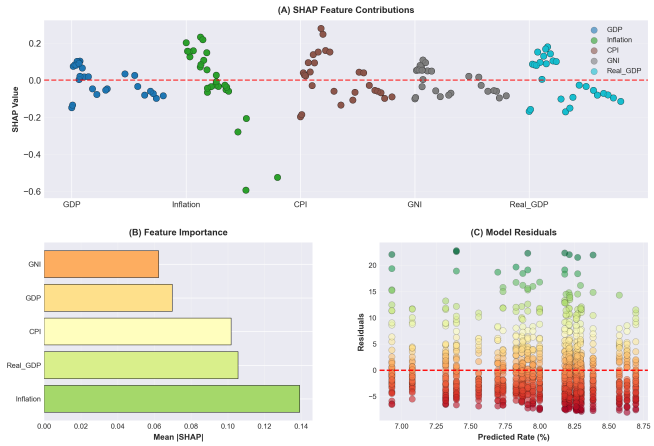

Fig. 4. SHAP summary. Panel (A): feature-wise contributions. Panel (B): mean absolute SHAP values ranking drivers. Panel (C): residuals.

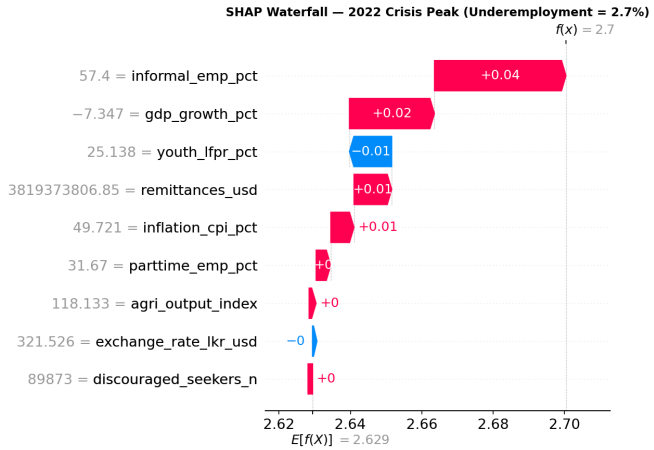

Fig. 5. SHAP waterfall for 2022 (crisis peak). GDP contraction and informal employment are the dominant amplifiers above baseline ϕ_0.

D. ARDL Bounds Test and ECM (RQ4)

The Pesaran F-bounds statistic is $\mathbf{F{=}7.25}$ (p=0.0008), exceeding the 1% I(1) upper bound of 4.68 for k=5 regressors (Case III) [19]. Cointegration is confirmed at 1%.

Long-run coefficients: GDP Growth ($\hat{\theta}$=−0.393), Inflation ($\hat{\theta}$=−0.142), Youth LFPR ($\hat{\theta}$=+0.260). A 1 pp GDP growth increase reduces long-run underemployment by 0.39 pp.

The ECM speed-of-adjustment $\hat{\lambda}=-1.087$ (p=0.003, 95% CI $[-1.81, -0.37]$) is negative and significant at 1%, indicating rapid but mildly overshooting quarterly correction consistent with volatile crisis-period dynamics. Diagnostics: Durbin-Watson $=$ 1.94; Breusch-Pagan p=0.11; Ljung-Box $Q(4)$ p=0.95; R^2=0.765.

E. Sensitivity and Robustness

A parallel XGBoost-SHAP analysis on the qualification-based underemployment series confirms Spearman rank correlation $\rho = 0.893$ ($p = 0.007$) across both definitions, exceeding the pre-registered ρ>0.6 threshold (Table III). Exchange Rate accounts for 75.8% of SHAP importance in the qualification-based model (vs. 34.0% hours-based), suggesting occupational mismatch is especially sensitive to currency depreciation.

TABLE III
SHAP FEATURE IMPORTANCE: HOURS-BASED VS QUALIFICATION-BASED

Feature	Hours %	Rank	Qual %	Rank	$\lvert\Delta\rvert$
Exchange Rate	34.0	1	75.8	1	0
GDP Growth	21.3	2	8.5	2	0
Youth LFPR	17.5	3	7.4	3	0
Agri. Output	14.8	4	1.1	5	1
Informal Emp.	10.0	5	0.6	6	1
Inflation†	1.5	6	6.4	4	2
Remittances	0.8	7	0.1	7	0

Spearman $\rho = 0.893$ ($p = 0.007$). †Rank shift ≥ 2.

Sub-period analysis (Table IV) shows moderate pre-crisis agreement ($\rho = 0.429$, $p = 0.337$); crisis and post-crisis sub-periods have n<5 and are illustrative only.

TABLE IV
SUB-PERIOD DEFINITION AGREEMENT (SPEARMAN ρ)

Period	n	ρ	p
Pre-crisis (2015–2019)	5	0.429	0.337
Crisis (2020–2022)	3	–	–
Post-crisis (2023–2024)	2	–	–
Overall	10	0.893	0.007

Sub-period rows with $n < 5$ are illustrative only; insufficient degrees of freedom for inference.

F. Gender-Disaggregated Analysis

Gender-disaggregated XGBoost-SHAP models reveal divergent driver structures (Table V): **Inflation** rises to rank 2 for female underemployment (33.3% SHAP) vs. rank 3 for males (16.8%), while **GDP Growth** is the second male driver (20.4%) but drops to rank 5 for females (4.2%).

The Wilcoxon signed-rank test confirms a statistically persistent female–male gap (W=55.0, p=0.001; Table VI). Mean gap: 1.23 pp; bootstrap 95% CI $[1.00, 1.40]$ during the crisis period. A Zivot-Andrews test on the gap series does not detect a significant break (t=−2.22), likely due to low power at n=10.

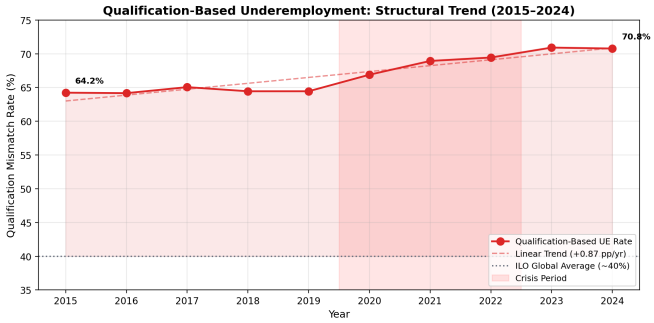

Fig. 6. Qualification-based underemployment trend (2015–2023). Structural rise of +0.87 pp/year with OLS trend line and 95% CI.

TABLE V
GENDER-DISAGGREGATED SHAP FEATURE RANKINGS

Feature	Agg %	Rank	Female %	Rank	Male %	Rank
Exchange Rate	49.4	1	41.4	1	48.5	1
Inflation	17.2	2	33.3	2	16.8	3
GDP Growth	10.6	3	4.2	5	20.4	2
Agri. Output	8.1	4	0.0	7	5.5	5
Youth LFPR	6.9	5	13.5	3	1.5	6
Remittances	4.0	6	3.0	6	1.2	7
Informal Emp.	3.8	7	4.6	4	6.1	4

Female–Male rank $\rho = 0.571$ ($p = 0.180$)

The qualification-based gender gap widens from 0.97 pp (2015) to 4.42 pp (2021; Fig. 7), indicating the crisis disproportionately affected female qualification mismatch.

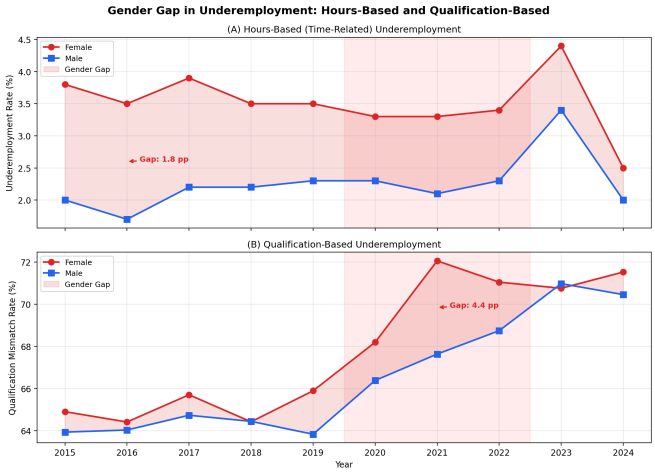

Fig. 7. Gender gap: hours-based (DCS) and qualification-based mismatch (2015–2023). Qualification gap widens from 0.97 pp to 4.42 pp.

G. District-Level Disaggregation

District-level analysis (25 districts, 2019–2022; Fig. 8) reveals substantial spatial heterogeneity. Cross-sectional regression of informal employment on underemployment yields weak associations (R^2=0.038 in 2019, R^2=0.059 in 2022; Table VII), indicating informality alone does not explain district-level variation.

TABLE VI
STATISTICAL TESTS ON THE GENDER GAP IN UNDEREMPLOYMENT

Test	Statistic	p	Result
Mean gap (F−M)	1.23 pp	–	Persistent female excess
Shapiro-Wilk (gap)	$W = 0.919$	0.351	Normal
Wilcoxon signed-rank	$W = 55.0$	0.0010	Significant
Bootstrap CI (crisis)	[1.00, 1.40]	–	Gap > 0 at 95%
ZA break (gap)	$t = -2.22$	–	Break: 2018 (–)

TABLE VII
DISTRICT-LEVEL INFORMAL EMPLOYMENT AND UNDEREMPLOYMENT REGRESSION

Year	n	Slope	R^2	p
2019	25	0.0279	0.038	0.3508
2022	25	0.0204	0.059	0.2414

V. POLICY RECOMMENDATIONS

Four evidence-based interventions target the SHAP-derived drivers.

Formalisation via social protection extension. EPF/ETF should be extended to informal workers (≈60% of the workforce) via a simplified contributory tier for micro-enterprises. ILO evidence [22] shows 15–20% smaller underemployment spikes in countries extending social protection during crises.

Youth qualification matching. Expansion of the Graduate Entrepreneurship Programme and a digital job-matching platform directly targets the Youth LFPR (rank 3 SHAP) channel. The World Bank Sri Lanka Human Capital Review [23] identifies TVET reform as the highest-priority structural intervention.

Remittance stabilisation. The 2021 remittance collapse from \$7.0 bn to \$5.5 bn preceded peak underemployment. When quarterly inflows fall below \$1.5 bn, an automatic subsidy from exchange equalisation reserves would partially replace this income buffer.

Adoption of ILO labour underutilisation framework. DCS should adopt the ILO LU1–LU4 indicators as quarterly headline statistics alongside unemployment, aligning with the 19th ICLS Resolution [5].

TABLE VIII
POLICY MAPPING TO SHAP-IDENTIFIED DRIVERS

Recommendation	Target Driver	Agency
EPF/ETF informal extension	Informal emp. (#2)	Labour Ministry
Graduate job platform	Youth LFPR (#3)	Higher Education
Remittance stabiliser	Remittances (lag)	CBSL
ILO LU1–LU4 framework	Measurement artifact	DCS

VI. CONCLUSION

Quarterly LFS micro-data reveals crisis underemployment peaks of 32.6%, nine times the 3.8% unemployment rate (**RQ1**). Structural breaks confirm 2021–2022 as a statistically significant rupture (**RQ3**), with macro breaks leading the labour market break by one year. SHAP establishes GDP contraction and informal employment as dominant drivers (**RQ2**), with Youth LFPR, remittances, and part-time employment capturing complementary channels. ARDL bounds testing (F=7.25) confirms long-run cointegration and resolves **RQ4**; the ECM confirms rapid quarterly correction ($\hat{\lambda}$=−1.087, p=0.003). Sensitivity analysis confirms consistency across definitions (ρ=0.893, p=0.007); gender analysis reveals a persistent female excess and widening qualification gap. District-level heterogeneity extends beyond informality. Aggregate unemployment is an inadequate crisis barometer for Sri Lanka.

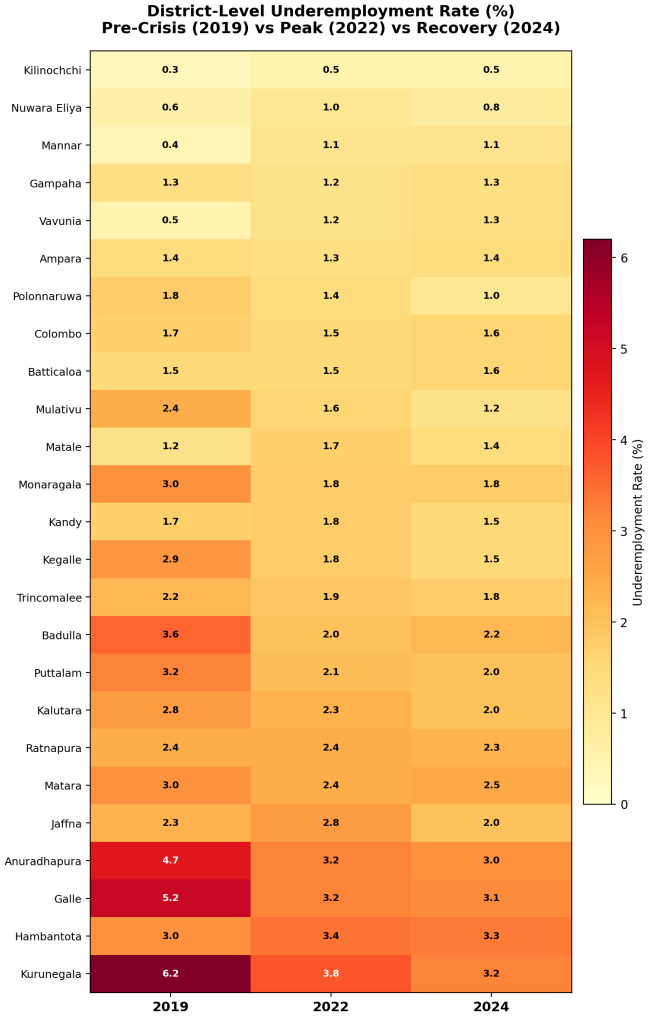

Fig. 8. District-level underemployment heatmap (25 districts, 2019/2022/2024). Northern and Southern districts show persistent hotspots.

Limitations. The annual XGBoost model (n=10) imposes degrees-of-freedom constraints; the quarterly panel mitigates this. SHAP captures predictive association, not causal direction. Real wages and sectoral employment shares were excluded due to insufficient quarterly availability. The 2022 LFS measurement artefact requires cautious interpretation. External validity may be limited to economies with comparable crisis-remittance constellations.

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